

## LORIENT BRETAGNE SUD AP, France

WMO: 072050

Lat: 47.7628N

Lon: 3.4356W

Elev: 49

StdP: 100.74

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -2.2       | -1.0 | -5.6                          | 2.4 | -0.3 | -3.9 | 2.7 | 0.6  | 14.8                  | 11.2 | 13.4 | 11.2 | 3.4                      | 60   | 0.585 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.7                    | 27.9            | 19.6 | 25.8 | 18.6 | 24.0 | 18.0 | 20.6                | 25.8 | 19.6 | 24.0 | 18.7 | 22.3 | 4.4                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.9                            | 13.7 | 22.1 | 18.1 | 13.1 | 21.0 | 17.4 | 12.5 | 20.2 | 59.2          | 25.8 | 55.8 | 24.1 | 53.2 | 22.3 | 25.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 11.7              | 10.3 | 9.2 | -4.6                       | 31.2 | 2.1                | 2.4 | -6.1                                               | 32.9 | -7.4       | 34.3 | -8.6       | 35.6 | -10.1      | 37.4 |  |  |
|                   |      |     | -5.3                       | 22.3 | 2.2                | 1.4 | -6.9                                               | 23.3 | -8.1       | 24.2 | -9.4       | 24.9 | -10.9      | 26.0 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.0   | 6.7   | 7.0   | 8.6   | 10.4  | 13.4  | 16.2  | 17.9  | 17.7  | 15.9  | 13.2  | 9.6   | 7.3  |
|                                                                                  | DBStd       | 4.97   | 3.37  | 2.99  | 2.64  | 2.69  | 2.86  | 2.85  | 2.61  | 2.56  | 2.49  | 2.72  | 2.91  | 3.53 |
|                                                                                  | HDD10.0     | 427    | 107   | 89    | 57    | 27    | 4     | 0     | 0     | 0     | 0     | 6     | 41    | 95   |
|                                                                                  | HDD18.3     | 2377   | 360   | 318   | 301   | 239   | 157   | 78    | 39    | 42    | 81    | 159   | 262   | 341  |
|                                                                                  | CDD10.0     | 1164   | 6     | 4     | 14    | 38    | 108   | 186   | 246   | 239   | 177   | 105   | 29    | 12   |
|                                                                                  | CDD18.3     | 73     | 0     | 0     | 0     | 0     | 3     | 13    | 27    | 23    | 7     | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 538    | 0     | 0     | 0     | 1     | 27    | 111   | 209   | 143   | 46    | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 116    | 0     | 0     | 0     | 0     | 2     | 25    | 47    | 37    | 5     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.5    | 5.0   | 4.8   | 4.7   | 4.6   | 4.4   | 4.2   | 4.2   | 3.9   | 4.0   | 4.3   | 4.5   | 4.9  |
| Precipitation                                                                    | PrecAvg     | 948    | 114   | 87    | 63    | 71    | 64    | 50    | 57    | 55    | 60    | 106   | 111   | 110  |
|                                                                                  | PrecMax     | 1184   | 236   | 218   | 191   | 172   | 136   | 113   | 116   | 140   | 152   | 199   | 229   | 217  |
|                                                                                  | PrecMin     | 716    | 16    | 15    | 4     | 21    | 12    | 7     | 6     | 9     | 0     | 15    | 25    | 22   |
|                                                                                  | PrecStd     | 126    | 61    | 51    | 42    | 42    | 30    | 30    | 26    | 33    | 42    | 42    | 52    | 54   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.2  | 14.3  | 19.0  | 22.4  | 26.2  | 29.8  | 30.7  | 30.3  | 27.3  | 21.8  | 16.7  | 14.4 |
|                                                                                  |             | MCWB   | 12.2  | 11.2  | 12.0  | 15.5  | 17.6  | 21.2  | 20.1  | 21.2  | 19.2  | 17.4  | 14.9  | 12.9 |
|                                                                                  | 2%          | DB     | 12.6  | 12.6  | 15.8  | 19.2  | 23.5  | 26.2  | 27.8  | 27.2  | 24.5  | 19.4  | 15.6  | 13.7 |
|                                                                                  |             | MCWB   | 11.7  | 10.6  | 11.2  | 13.4  | 16.9  | 19.0  | 19.3  | 19.5  | 17.7  | 16.3  | 14.1  | 12.6 |
|                                                                                  | 5%          | DB     | 12.1  | 11.9  | 13.9  | 17.1  | 20.8  | 24.0  | 25.7  | 24.4  | 22.3  | 18.2  | 14.9  | 13.1 |
|                                                                                  |             | MCWB   | 11.1  | 10.4  | 10.6  | 12.4  | 15.5  | 17.9  | 18.7  | 18.7  | 16.9  | 15.8  | 13.6  | 12.2 |
|                                                                                  | 10%         | DB     | 11.5  | 11.2  | 12.6  | 15.1  | 18.5  | 21.8  | 23.5  | 22.5  | 20.5  | 17.2  | 14.2  | 12.5 |
|                                                                                  |             | MCWB   | 10.6  | 10.0  | 10.3  | 11.5  | 14.4  | 16.7  | 17.9  | 17.9  | 16.4  | 15.3  | 12.8  | 11.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.7  | 12.1  | 13.2  | 16.2  | 19.1  | 22.0  | 21.2  | 22.1  | 20.3  | 18.2  | 15.6  | 13.6 |
|                                                                                  |             | MCDB   | 13.0  | 13.0  | 16.7  | 20.5  | 24.6  | 28.0  | 27.7  | 28.7  | 25.0  | 20.0  | 16.1  | 13.9 |
|                                                                                  | 2%          | WB     | 12.1  | 11.4  | 12.2  | 14.2  | 17.3  | 19.8  | 20.1  | 20.5  | 19.0  | 17.3  | 14.6  | 13.0 |
|                                                                                  |             | MCDB   | 12.4  | 12.1  | 14.2  | 17.9  | 21.6  | 25.1  | 25.9  | 25.0  | 21.9  | 18.6  | 15.3  | 13.5 |
|                                                                                  | 5%          | WB     | 11.5  | 10.9  | 11.6  | 13.0  | 16.1  | 18.4  | 19.2  | 19.5  | 18.1  | 16.5  | 14.0  | 12.5 |
|                                                                                  |             | MCDB   | 11.9  | 11.6  | 13.1  | 15.9  | 20.0  | 22.8  | 23.9  | 22.9  | 20.5  | 17.7  | 14.7  | 13.0 |
|                                                                                  | 10%         | WB     | 10.8  | 10.4  | 11.0  | 12.1  | 14.9  | 17.2  | 18.4  | 18.6  | 17.3  | 15.8  | 13.1  | 11.8 |
|                                                                                  |             | MCDB   | 11.4  | 11.1  | 12.1  | 14.3  | 17.8  | 20.8  | 22.3  | 21.4  | 19.5  | 16.9  | 14.0  | 12.4 |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 5.6   | 6.4   | 7.4   | 8.4   | 8.4   | 8.8   | 8.7   | 8.9   | 9.0   | 7.2   | 6.5   | 5.7  |
|                                                                                  |             | MCDBR  | 4.8   | 6.3   | 9.5   | 11.4  | 11.9  | 12.4  | 12.8  | 11.8  | 11.4  | 7.8   | 6.1   | 4.5  |
|                                                                                  |             | MCWBR  | 4.5   | 4.9   | 6.1   | 6.7   | 6.6   | 6.6   | 6.1   | 5.7   | 5.7   | 5.2   | 5.0   | 4.4  |
|                                                                                  | 5% WB       | MCDBR  | 4.6   | 5.4   | 7.1   | 9.2   | 10.4  | 11.1  | 10.9  | 9.9   | 8.7   | 6.4   | 5.1   | 4.5  |
|                                                                                  |             | MCWBR  | 4.7   | 4.8   | 5.1   | 5.9   | 6.2   | 6.4   | 5.8   | 5.5   | 5.3   | 4.9   | 4.8   | 4.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333  | 0.346 | 0.372 | 0.384 | 0.389 | 0.386 | 0.383 | 0.373 | 0.360 | 0.359 | 0.338 | 0.326 |      |
|                                                                                  | taud        | 2.417  | 2.396 | 2.337 | 2.317 | 2.349 | 2.384 | 2.393 | 2.441 | 2.458 | 2.455 | 2.438 | 2.395 |      |
|                                                                                  | Ebn at Noon | 736    | 809   | 842   | 867   | 875   | 877   | 874   | 870   | 848   | 785   | 724   | 691   |      |
|                                                                                  | Edh at Noon | 69     | 86    | 106   | 119   | 119   | 116   | 114   | 104   | 93    | 80    | 67    | 63    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.04   | 1.94  | 3.12  | 4.69  | 5.55  | 6.10  | 5.83  | 5.06  | 4.02  | 2.29  | 1.31  | 0.90  |      |
|                                                                                  | RadStd      | 0.10   | 0.20  | 0.31  | 0.46  | 0.38  | 0.57  | 0.28  | 0.35  | 0.28  | 0.19  | 0.11  | 0.10  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page